

Freight Rate Prediction

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Overview

I trained on 48,000 labeled loads from January to October 2025 to predict 12,000 loads from November and December 2025. The final model is LightGBM. On a held-out test set of the last two months it has a mean absolute error of \$56.4 per load on normal loads, compared with \$92.0 for a baseline that multiplies the typical rate per mile for the equipment and distance band by the distance.

Data cleaning

Weights of zero or less were treated as missing. Missing weights (300 labeled, 165 validation, plus 292 and 145 negative) were filled with the median weight for the same equipment type, learned from the training data. Missing market index values (374 labeled, 249 validation) were filled with the median market index of other loads on the same date, since the index is a daily market measure. The other approach beside this was to transform negative values to positive with absolute value.

About 1.4% of the rates were corrupted. Compared with similar loads (same equipment and distance band), normal rates sit between 0.81x and 1.34x the typical rate, while these sit below 0.47x or above 2.14x, with nothing in between, and no feature explains them. The code places the cutoffs in the middle of these two gaps and removes the loads from the training data only.

Eight cities in validation never appear in the labeled data (about 12% of validation loads), so cities are represented by their coordinates instead of their names.

Split and validation

Validation comes after all labeled data in time, so I split by date rather than randomly to prevent data leakage. A random split would put loads from the same days in both sets and overstate accuracy. All fill values and outlier cutoffs are learned from the training part only, and validation rows are never dropped.

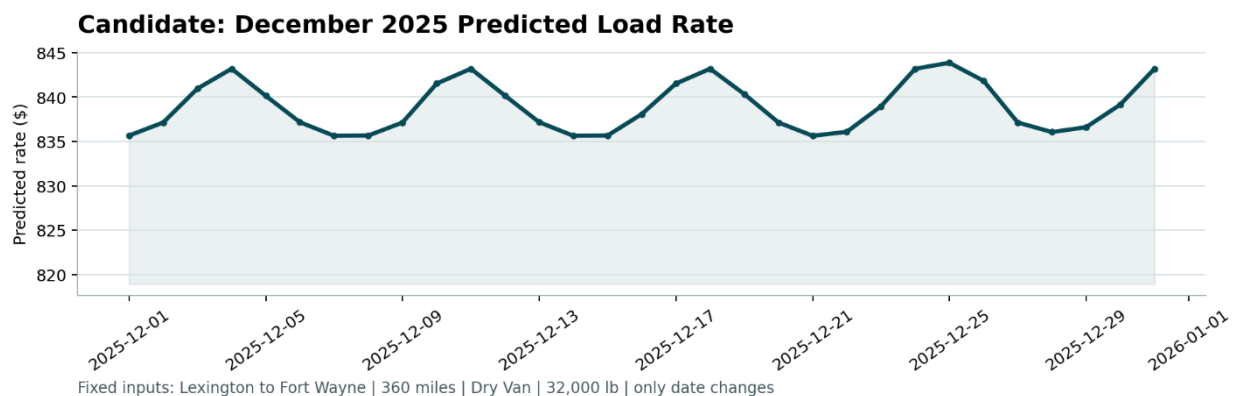
I tuned four models with randomized search and time-series cross-validation inside the training period, chose the one with the lowest cross-validation error, and then evaluated on the test set once. The final model was retrained with the same settings on all labeled data before predicting validation.

Results

Cross-validation MAE was \$73.6 for LightGBM, \$78.6 for XGBoost, \$81.7 for Random Forest and \$86.4 for scikit-learn gradient boosting. On test loads without corrupted rates, the MAE was \$56.4 for LightGBM, \$56.0 for Random Forest, \$56.3 for gradient boosting and \$60.4 for XGBoost, against \$92.0 for the baseline. LightGBM also had the lowest RMSE (\$83.7).

December chart

The December file has no coordinates, market index or quote signal. Coordinates were looked up by city, each day's market index was taken from the median of the validation loads on that date, and the quote signal was set to the training median. With the load fixed, the model predicts \$835.64 to \$843.87 (about \$2.33 per mile), following a weekly pattern from the market.



Limitations

The labeled data ends in October, so the model has no knowledge of holidays and treats November and December like October. The feature comparisons were made on the test set, so the reported test error may be slightly optimistic.